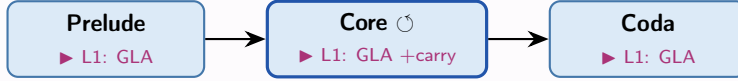


Retention — Gated Linear Attention Memory

a recurrent memory branch in parallel to attention

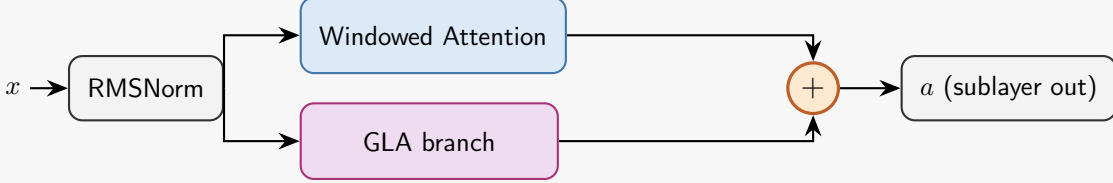
Placement



GLA branch on layer 1 of **all three** sections • cross-iteration *carry* (Axis 2) is **core-only**

Integration (layer 1 of prelude / core / coda)

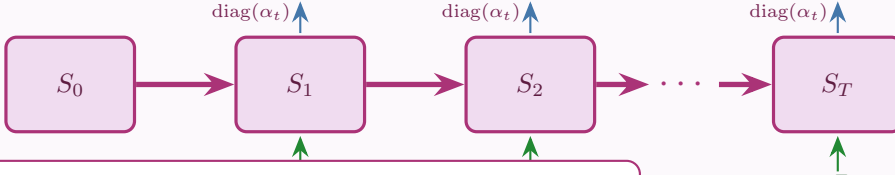
$$a = \text{attn}(x) + \sigma(g) \cdot \text{GLA}(\text{norm}(x), S)$$



gate $\sigma(g) \approx 0$ at init $\Rightarrow$ branch off (identity to baseline)

Axis 1 — recurrence over the sequence (the SSM hidden state)

one layer application scans tokens $t = 1 \dots S$, evolving the matrix state $S \in \mathbb{R}^{d_k \times d_v}$ per head



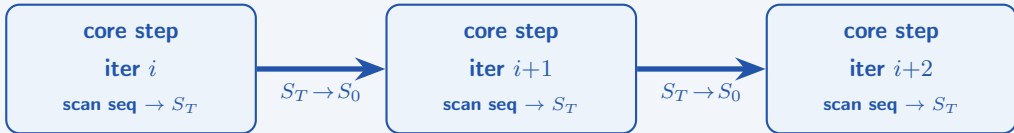
$$S_t = \text{diag}(\alpha_t) S_{t-1} + k_t^T v_t, \quad o_t = q_t S_t$$

forget gate $\alpha_t = \sigma(x_t W_g) \in (0, 1)^{d_k}$ (data-dependent, per key channel)

kernel: chunked-parallel scan
fp32 state; $b = \text{cumsum}(\log \alpha)$ clamp ≥ -30

Axis 2 — recurrence over the loop (the cross-iteration carry, retention_carry)

core only: the final sequence state S_T of loop iter i seeds the initial state of iter $i+1$ — memory across thinking-depth



no-grad iterations carry a *detached* state (truncated BPTT, last 4 iters) • `retention_carry=false` $\Rightarrow$ reset each iter (Axis 1 only)

Two recurrences, one is standard SSM. Axis 1 (over the sequence) is the linear-attention/SSM hidden state — always on. Axis 2 (over the loop) is MORPH-specific and *optional*: it adds a second recurrence over thinking-depth on top. `carry=false` = pure per-iteration SSM; `carry=true` = +depth memory. The branch runs *in parallel* to attention (gated sum) — never recurrent over attention.